



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO K.A.S. AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

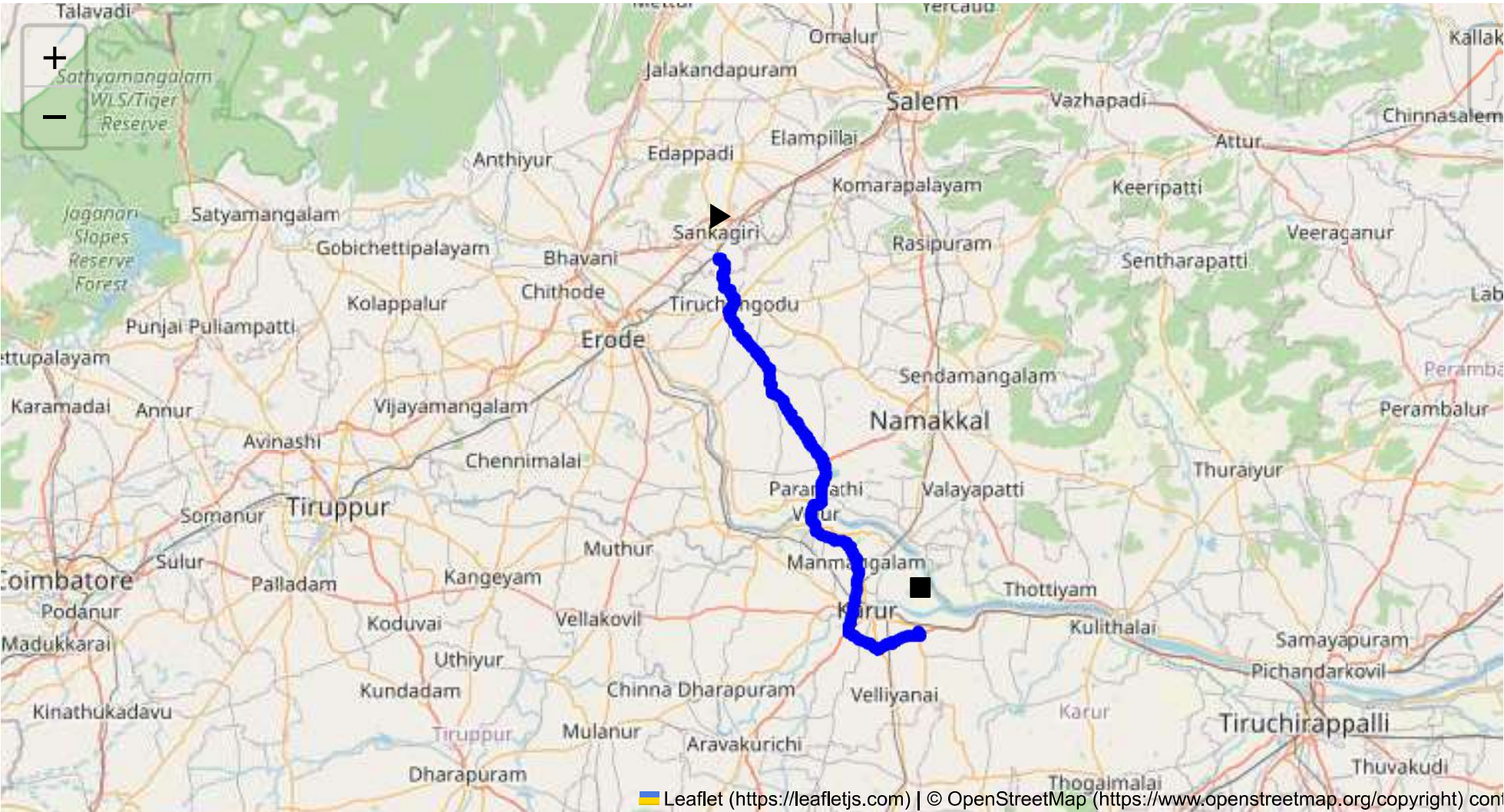
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 80.69 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (10.930584, 78.152246)



Welcome to the Journey Risk Management Study

Route Overview

The route between Sangagiri and A.P. Nagar in Tamil Nadu is approximately 80.69 kilometers and usually takes about 1.53 hours for heavy vehicles. This path traverses through largely rural areas with some stretches near urban centers. Major roads on this route might include state highways and district roads connecting small towns and villages.

Weather Conditions

Tamil Nadu experiences tropical weather, characterized by high temperatures and humidity. The region has three distinct seasons: summer (March to June), monsoon (June to September), and winter (December to February). During the monsoon, heavy rainfall can lead to waterlogging and slippery roads, increasing the risk of hydroplaning and accidents. Plan for potential delays during monsoon months.

Traffic Patterns

Traffic can be moderate to heavy near urban centers and junctions, notably during peak commuting hours in the morning (8-10 AM) and evening (5-7 PM). Rural sections generally experience lighter traffic, but slow-moving agricultural vehicles can be prevalent.

Road Quality and Infrastructure

Roads in this area are generally maintained but can vary significantly, with sporadic potholes and uneven surfaces particularly after the monsoon. Narrow sections, especially in rural areas, could be challenging for heavy vehicles. Watch out for pedestrians and livestock.

Alternative Routes

In emergencies, alternative routes might include bypassing certain towns or using rural roads parallel to main highways. However, these alternatives should be carefully evaluated for road conditions and accessibility, as not all are suitable for heavy vehicles.

Regulations on Hazardous Materials

Tamil Nadu imposes specific regulations on transporting hazardous materials, including designated routes and time restrictions. Ensure compliance with the Transport Department's guidelines and have the necessary permits and documentation readily available.

Historical Incidents

This region has witnessed vehicle accidents involving heavy trucks, often due to poor road conditions or driver error. There is limited data on incidents specifically involving hazardous materials, but trucks must remain cautious, particularly on narrow and winding roads.

Environmental Considerations

The route includes rural landscapes and may pass near agricultural lands and water bodies. Avoid contamination and leakage of hazardous substances to protect these sensitive areas. Compliance with environmental regulations is imperative.

Communication Coverage

The route likely has inconsistent mobile network coverage, particularly in rural sections. Anticipate dead zones that might interrupt communication; ensure vehicles are equipped with reliable communication

devices such as satellite phones or radios.

Emergency Response Times

Emergency response in rural areas can be slower, possibly exceeding 30 minutes depending on the distance to the nearest town or hospital. Urban areas along the way may offer quicker response times. It is crucial to have emergency contacts and clear procedures in place.

Overall Risk Assessment

While the route is manageable, several risk factors need attention: potential weather impacts, variable road quality, inconsistent traffic conditions, and limited communication in rural areas. Proactive measures, including thorough route planning, driver training, and regular vehicle checks, are recommended to mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
10	Blind Spot	Blind Spot	10.92515, 78.15309	10 KM/Hr
11	Turn	High	10.93062, 78.15246	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium